Project: Worked examples
Title: Shaft under combined bending and torsion
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Bending and torsional stresses in a solid round shaft with stress-concentration factors, the maximum shear (Tresca) and von Mises equivalent stresses, and the factors of safety against yield, against a required factor.

WHAT IT CALCULATES	RELATION
Bending stress, with its stress concentration	$\sigma = K_t \cdot 32M/(\pi d^3)$
Torsional shear stress, with its stress concentration	$\tau = K_{ts} \cdot 16T/(\pi d^3)$
Maximum shear stress (Tresca)	$\tau_{max} = \sqrt{((\sigma/2)^2 + \tau^2)}$
von Mises equivalent stress	$\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$
Factors of safety against yield, the lower governing	$n_T = (S_y/2)/\tau_{max}$; $n_{VM} = S_y/\sigma'$

Basis: Maximum-shear-stress and distortion-energy theories[1](§5-4, §5-5); Shaft design for stress[1](§7-4).

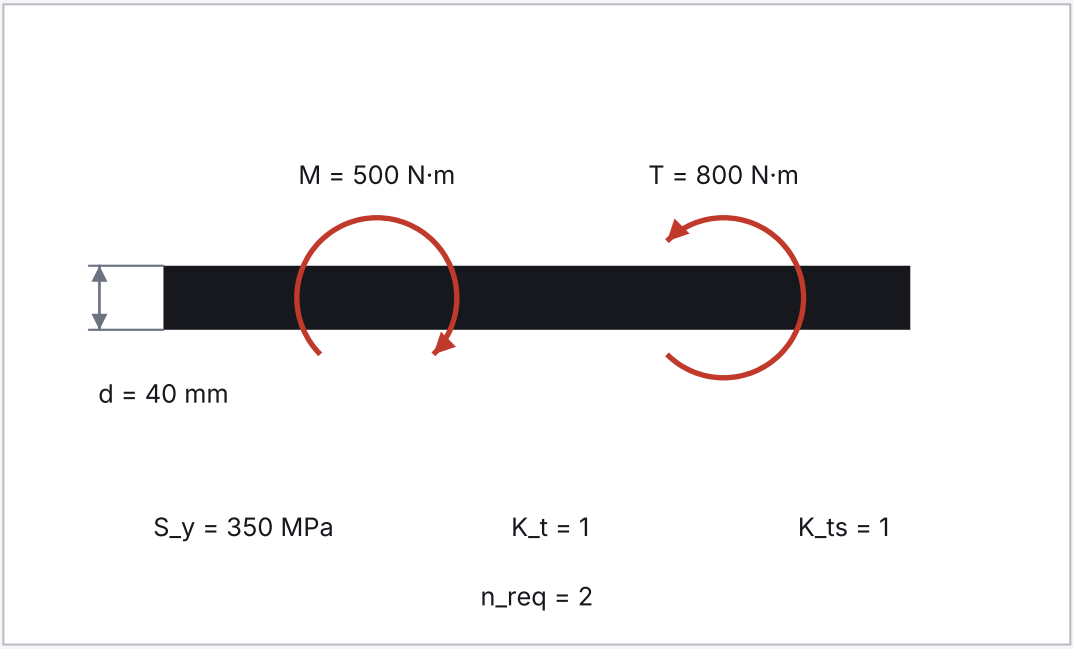


Figure 1 — A solid round shaft under bending and torsion at the section. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
d	Shaft diameter	40	mm	drawing: DRG-SH-401 rev A	1 … 2000 mm
M	Bending moment (magnitude)	500	N·m	spec: LOAD-SCHEDULE-01	0 … 10000000 N·m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
T	Torque (magnitude)	800	N·m	spec: LOAD-SCHEDULE-01	0 … 10000000 N·m
S_y	Yield strength	350	MPa	drawing: DRG-SH-401 rev A	50 … 3000 MPa
K_t	Bending stress-concentration factor (1 for none)	1		drawing: DRG-SH-401 rev A	1 … 5
K_ts	Torsional stress-concentration factor (1 for none)	1		drawing: DRG-SH-401 rev A	1 … 5
n_req	Required factor of safety	2		spec: SPEC-MECH-03 design criteria	1 … 10

3. Assumptions

A1. Solid round section, linear elastic; static loading. (Fatigue is mech.fatigue-miner's job; hollow shafts are not covered.)

4. Working

```
[8] sigma = K_t * 32 * M / (pi * d^3)                                    [1](§7-4)
= 1 × 32 × 500 N·m / (π × (40 mm)^3)
= 79.577 MPa
```
Bending stress

```
[9] tau = K_ts * 16 * T / (pi * d^3)                                     [1](§7-4)
= 1 × 16 × 800 N·m / (π × (40 mm)^3)
= 63.662 MPa
```
Torsional shear stress

```
[10] tau_max = sqrt((sigma / 2)^2 + tau^2)                               [1](§5-4)
= sqrt((79.577 MPa / 2)^2 + (63.662 MPa)^2)
= 75.073 MPa
```
Maximum shear stress (Tresca)

```
[11] sigma_vm = sqrt(sigma^2 + 3 * tau^2)                                [1](§5-5)
= sqrt((79.577 MPa)^2 + 3 × (63.662 MPa)^2)
= 135.98 MPa
```
von Mises stress

```
[12] n_T = (S_y / 2) / tau_max
= 350 MPa / 2 / 75.073 MPa
= 2.3311
```
Factor of safety, maximum shear stress

```
[13] n_VM = S_y / sigma_vm
= 350 MPa / 135.98 MPa
= 2.5739
```
Factor of safety, distortion energy

```
[14] n_min = min(n_T, n_VM)
= min(2.3311, 2.5739)
= 2.3311
```
Governing factor of safety

```
[15] u = n_req / n_min
= 2 / 2.3311
= 0.85798
```
Required over governing factor of safety

5. Results

SYMBOL	VALUE	DESCRIPTION
sigma	79.577 MPa	Bending stress
tau	63.662 MPa	Torsional shear
tau_max	75.073 MPa	Maximum shear stress
sigma_vm	135.98 MPa	von Mises stress
n_T	2.3311	Factor of safety (Tresca)
n_VM	2.5739	Factor of safety (von Mises)
n_min	2.3311	Governing factor of safety

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Factor of safety against the required factor	u = 0.85798	u ≤ 1	0.858	[1](§7-4)

7. Applicability

No warnings.

8. Verification

Template mech.shaft-stress@1.0.1 Content hash 4da066e367477095 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.